

MANOUBA SCHOOL OF ENGINEERING

BUILDING FOOTPRINT TRACKER

PRESENTATION OVERVIEW

1. Introduction
2. Area of Interest
3. Business Model Canvas
4. Use Case Diagram
5. Product Backlog
6. Release Plan
7. Conclusion
8. Q&A



INTRODUCTION

Urban expansion in Tunisia is accelerating at an unprecedented pace, and this trend is especially critical around **Sebkhat Séjoumi** an ecologically fragile, Ramsar listed wetland increasingly threatened by informal and uncontrolled construction. To respond to these pressures, our Building Footprint Tracker offers an AI-driven, web-based GIS platform capable of automatically detecting and monitoring building footprints from satellite imagery in real time. By delivering precise, continuously updated insights on urban growth, the system empowers urban planners, environmental agencies, researchers, and local authorities with the data they need to better manage *flood risks, regulate illegal expansion, safeguard biodiversity, reduce pollution, and guide sustainable infrastructure* planning across one of Greater Tunis's most vulnerable landscapes.

REGION COVERED



BUSINESS MODEL CANVAS

Business Model Canvas

0

Key Partners
- Satellite imagery providers (NASA, ESA, Google Earth Engine)
- GIS & AI/ML research labs and experts
- Cloud service providers (AWS, Azure, Google Cloud)
- Universities and urban planning institutions
- Government agencies and municipalities

Key Activities
- Collecting and preprocessing satellite imagery
- Developing and maintaining the GIS platform
- Training AI models for building detection
- Updating and validating footprint datasets
- Integrating periodic or real-time updates

Key Resources
- Satellite imagery datasets
- Trained AI/ML models for footprint extraction
- Cloud infrastructure and geospatial servers
- Web GIS interface (frontend and backend)
- Skilled data science and GIS development team

Value Propositions
- Automated detection and mapping of buildings from satellite images
- Accurate monitoring of urban growth and land-use change
- Data-driven insights for sustainable urban planning
- Visualization of temporal changes in urban environments
- Accessible and scalable web-based geospatial service

Customer Relationships
- Self-service web platform with dashboards
- Technical support and user documentation
- Research and data-sharing collaborations
- Institutional partnerships and co-development

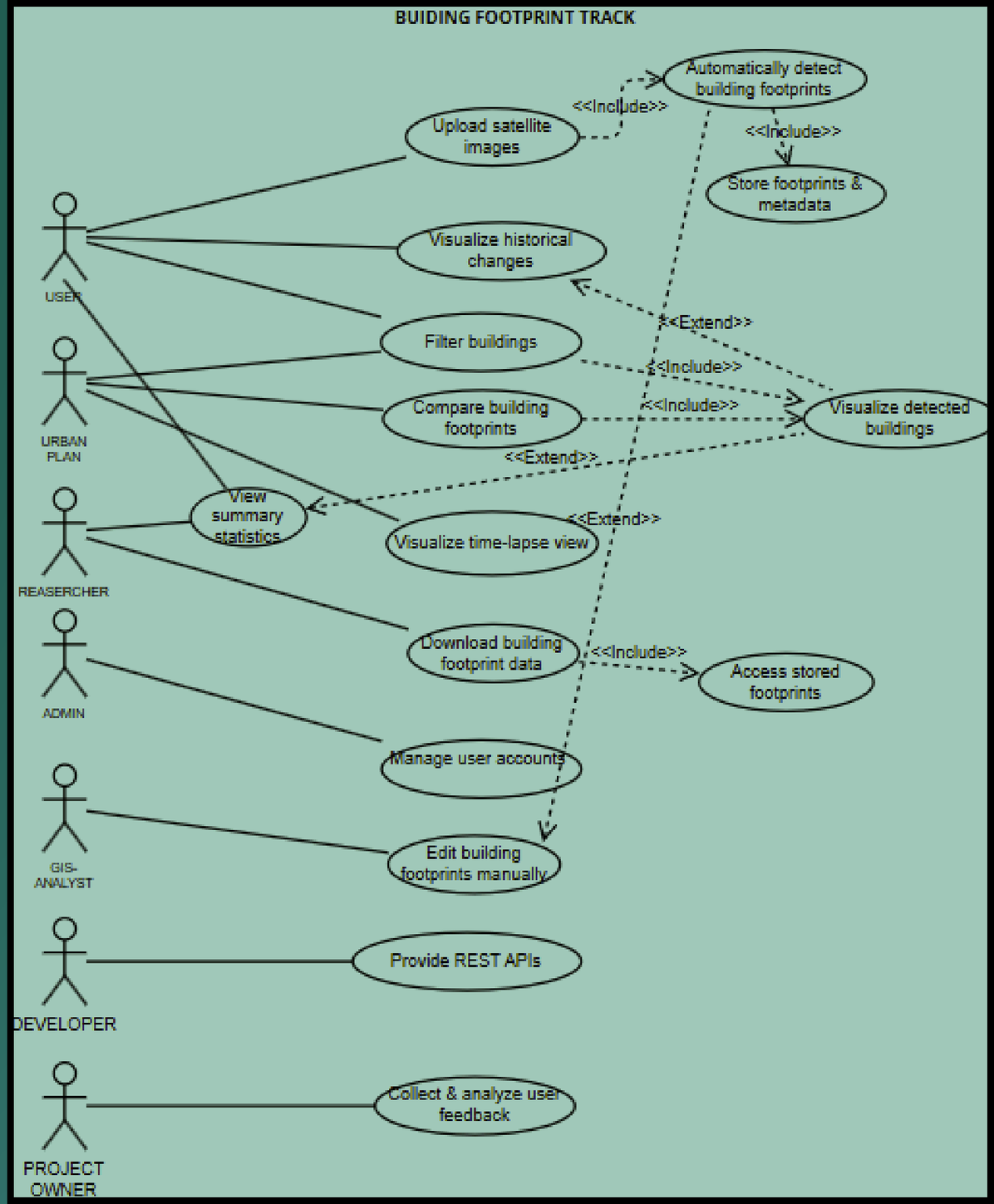
Channels
- Web application portal
- API access for integration with other platforms
- Research collaborations and data partnerships
- Social media and academic conferences for outreach

Customer Segments
- Urban planners and local authorities
- Environmental researchers
- Real estate developers
- NGOs and sustainability organizations
- Academic and research institutions

Cost Structure
- Satellite imagery acquisition and processing
- Cloud hosting and infrastructure costs
- AI model training and maintenance
- Platform development and staff salaries
- Marketing and institutional collaboration expenses

Revenue Streams
- Subscription-based access (monthly/yearly)
- Paid custom analytics and data visualization services
- Consulting for urban and environmental monitoring
- Government or NGO partnerships and grants
- API usage fees for developers

USE CASE DIAGRAM



PRODUCT BACKLOG

Building Footprint Tracker – Product Backlog

As a	I want to	So that	Priority
User	Visualize historical changes through	I can better understand the evolution of urban growth	Low
System	Receive notifications when new sateilite	I can update my analyses promptly	Low
System	Store all footprints and metadata in a	Data remains consistent and accessible	High
Developer	Provide RESTful APIs for the system	Other applications can integrate with the Building Footprint Tracker	Medium
User	Use the application on different devices	It's accessible to a wider audience	Low
Project Owner	Monitor system performance and	I can ensure the platform runs efficiently	Low
Project Owner	Collect and analyze user feedback	The platform can be improved for future releases	Medium

PRODUCT BACKLOG

Building Footprint Tracker - Product Backlog

As a	I want to	So that	Priority
User	Upload satellite images	I can analyze new regions and extract building data	High
System	Automatically detect building footprints from satellite imagery	Users can save time and avoid manual mapping	High
User	Visualize detected buildings on an interactive map	I can easily explore spatial patterns and results	High
Urban Planner	Compare building footprints from different time periods	I can monitor urban expansion and land use change	High
Researcher	Download building footprint data as GeoJSON or shapefiles	I can perform further GIE or statistical analysis	Medium
User	View summary statistics (number of buildings, total areal	I can quickly understand he scale of urban developme	Medium
User	Filter buildings by area size, location, or date	I can focus on specific subsets of data	Medium
GIS Analyst	Edit or correct detected building footprints manually	The dataset becomes more accurate	Low
Admin	Manage user accounts and roles	I can control who uploads, edits, or downloads data	Medium
User	Visualize historical changes through time-lapse animation	I can better understand the evolution of urban	Low
System	Receive notifications when new satellite images are available	I can update my analyses promptly	Low
Developer	Store all footprints and metadata in a spatial database (PostG(S)	Data remains consistent and accessible	High
	Provide RESTful APIs	Other applications	

Week	Milestone	Key Deliverables
Week 1	Platform Skeleton Functional	•Working dev environment • Basic map with satellite imagery • User authentication (login/signup) • Database schema finalized
Week 2	Core Detection Capability Working	• Building detection on sample areas • Map showing detected footprints • Basic export (GeoJSON)
Week 3	MVP Feature-Complete	• Change detection working • User-friendly interface • Analytics visualization • Quick-start documentation
Week 4	Platform Ready for External Users	• Stable, polished platform • Beta onboarding kit • Demo regions ready
Week 5	Platform Launched & Validated	• Live platform with active users • Launch announcement published

CONCLUSION

- Web-based GIS tool to track building footprints from satellite imagery.
- Helps monitor urban growth and land-use change.
- Supports sustainable urban planning and data-driven decisions.
- Clear roadmap from concept to implementation.





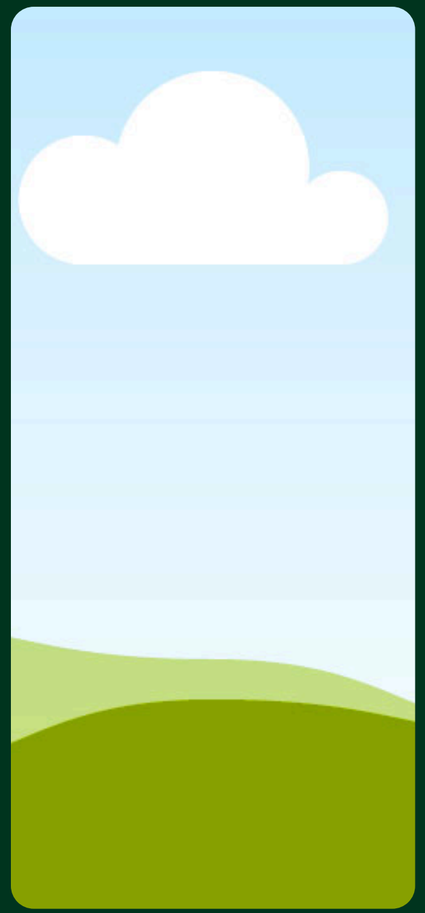
Presenter: Avery Davis, Graduate Researcher

THANK YOU FOR YOUR ATTENTION AND YOUR TIME

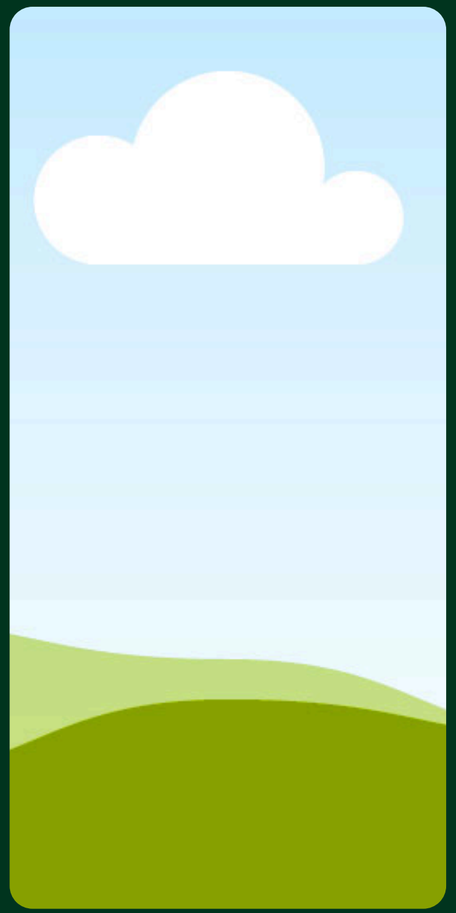
A Research Study Presented at the Academic Futures Conference 2029

Institution: Borcelle University – Department of Environmental Studies

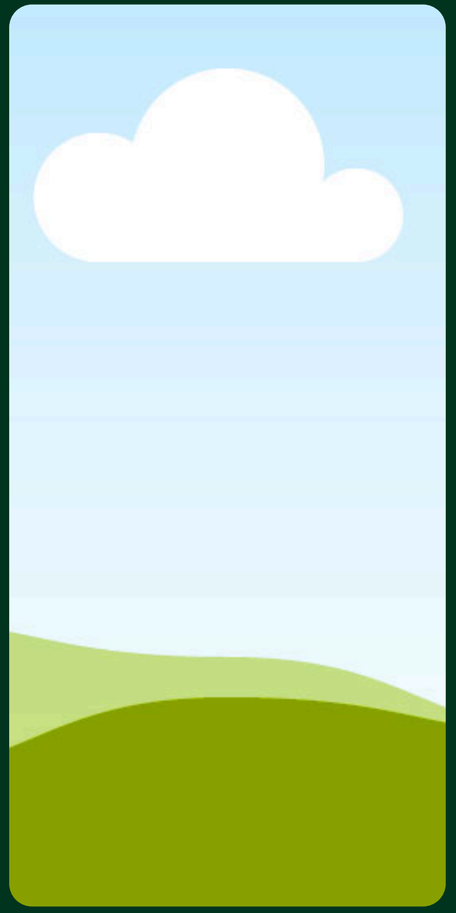
OUR TEAM



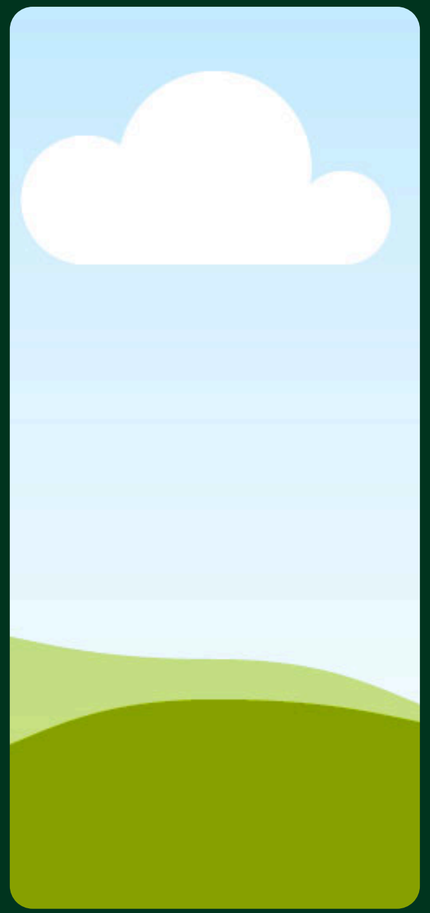
Balkis Selmi



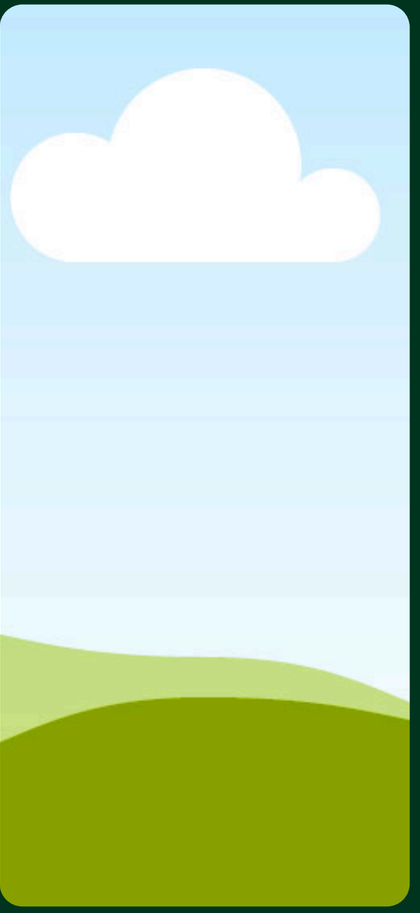
Farah Ben Choug



Ghofrane Dabbebi



Nour Elimen Sbaiti



Zayneb Ben Rajab